Cyber Security Project

Part A — Active Directory Infrastructure & Privilege Escalation

Student:	Adam
Domain:	adamklab.local
Environment:	VMware Workstation — isolated local lab
Machines:	DC01 (Windows Server 2022) • Win11-Client (Windows 11 Pro)
Submission deadline:	20.11.2026

This report documents Part A of the project: a Windows Server 2022 Active Directory domain (**adamklab.local**) was built and a Windows 11 workstation was joined to it. A realistic Group Policy misconfiguration was then introduced and exploited, escalating a standard domain user to local Administrator on the workstation.

1. Lab Architecture

The lab consists of two virtual machines on the same isolated network: a Domain Controller running Active Directory and DNS, and a Windows 11 workstation joined to the resulting domain.

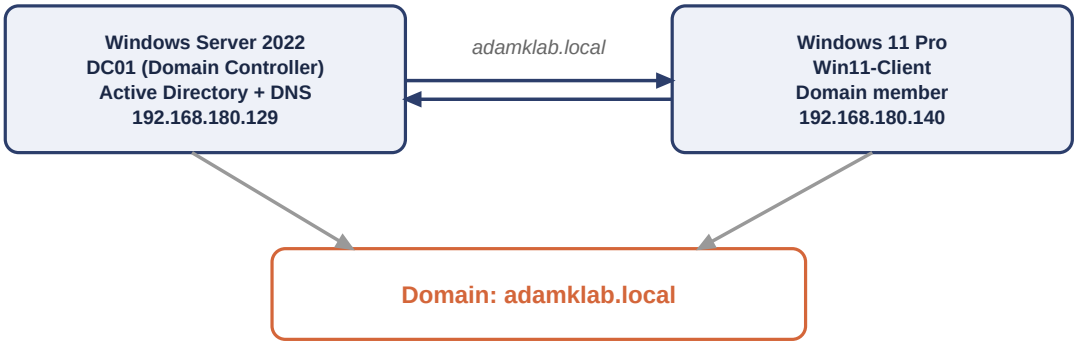


Figure 1 — Lab topology: Domain Controller and domain-joined workstation.

2. Active Directory Domain — Operational

Windows Server 2022 was promoted to a Domain Controller, creating the new forest/domain **adamklab.local**, with AD DS and DNS running on a static IP address. The server is shown below as healthy and reachable, with the new domain visible in Active Directory Users and Computers.

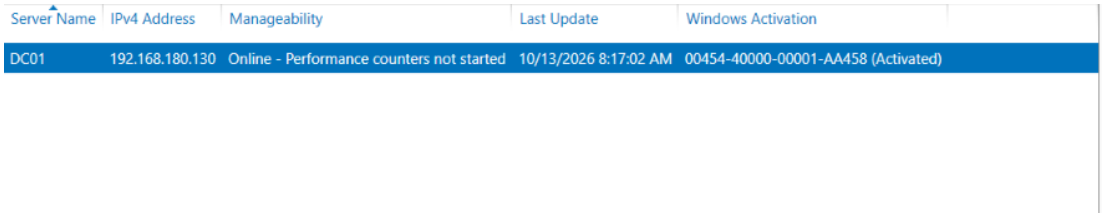


Figure 2 — DC01 listed as Online with a static IP and an activated license.

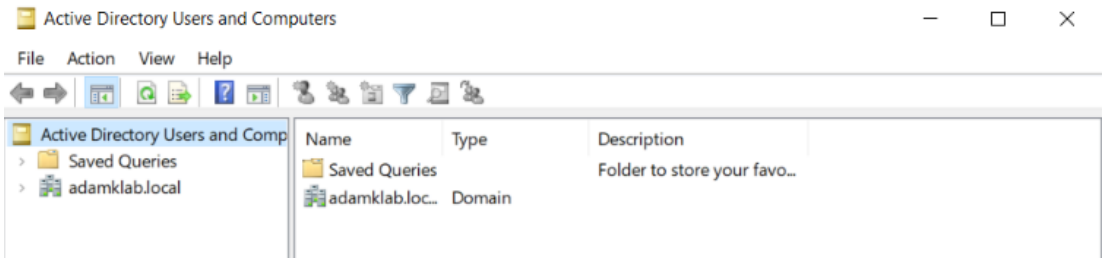


Figure 3 — The 'adamklab.local' domain visible in Active Directory Users and Computers.

3. Windows 11 Workstation Joined to the Domain

A Windows 11 Pro workstation (**Win11-Client**) was built and joined to adamklab.local. Domain authentication succeeds directly on the workstation, confirming the join was successful.

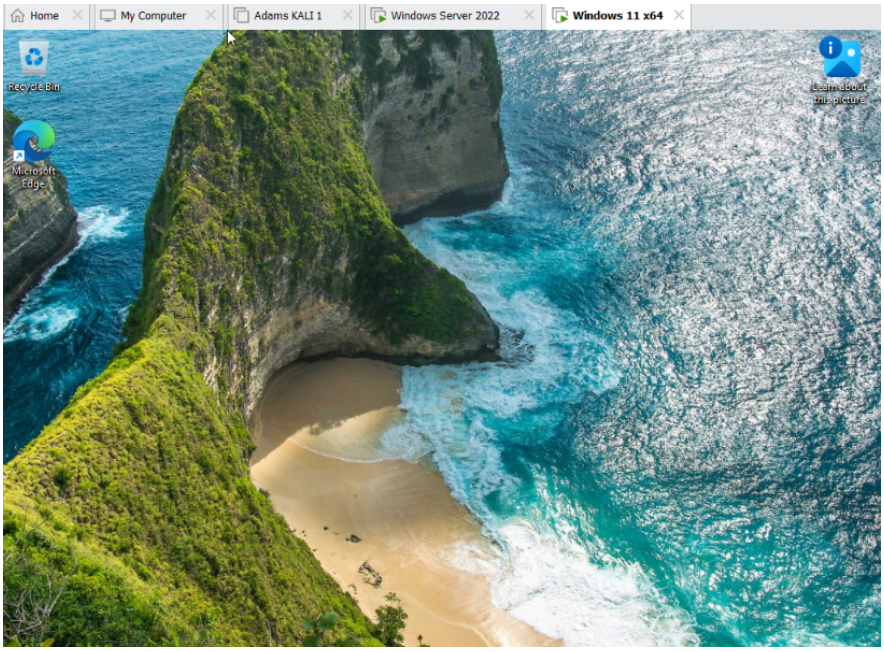


Figure 4 — Successful domain login on Win11-Client as adamklab\administrator.

A standard, non-privileged domain user — **Student1** — was then created and used as the baseline account for the privilege-escalation test below. Its group membership (`whoami /groups`) initially showed no administrative rights.

```
Command Prompt

C:\Users\Student1>whoami /groups

GROUP INFORMATION
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Group Name                                 Type             SID          Attributes
========================================== ================ ============ ==================================================
Everyone                                   Well-known group S-1-1-0      Mandatory group, Enabled by default,
Enabled group
BUILTIN\Users                              Alias            S-1-5-32-545 Mandatory group, Enabled by default,
Enabled group
NT AUTHORITY\INTERACTIVE                   Well-known group S-1-5-4      Mandatory group, Enabled by default,
Enabled group
CONSOLE LOGON                              Well-known group S-1-2-1      Mandatory group, Enabled by default,
Enabled group
NT AUTHORITY\Authenticated Users           Well-known group S-1-5-11     Mandatory group, Enabled by default,
Enabled group
NT AUTHORITY\This Organization             Well-known group S-1-5-15     Mandatory group, Enabled by default,
Enabled group
LOCAL                                      Well-known group S-1-2-0      Mandatory group, Enabled by default,
Enabled group
Authentication authority asserted identity Well-known group S-1-18-1     Mandatory group, Enabled by default,
Enabled group
Mandatory Label\Medium Mandatory Level     Label            S-1-16-8192


C:\Users\Student1>
```

Figure 5 — whoami /groups as Student1 — no BUILTIN\Administrators membership (baseline).

4. Exploiting a Group Policy Misconfiguration

The core of the assignment was to discover and exploit a misconfiguration granting a standard user local Administrator rights. A realistic, commonly-seen misconfiguration was used: an overly-permissive **'Restricted Groups'** setting in the Default Domain Policy, which adds a standard user directly into the local **Administrators** group of every computer in the domain. This mirrors a real-world incident where one domain-wide policy change silently grants excessive local privileges across an entire organization.

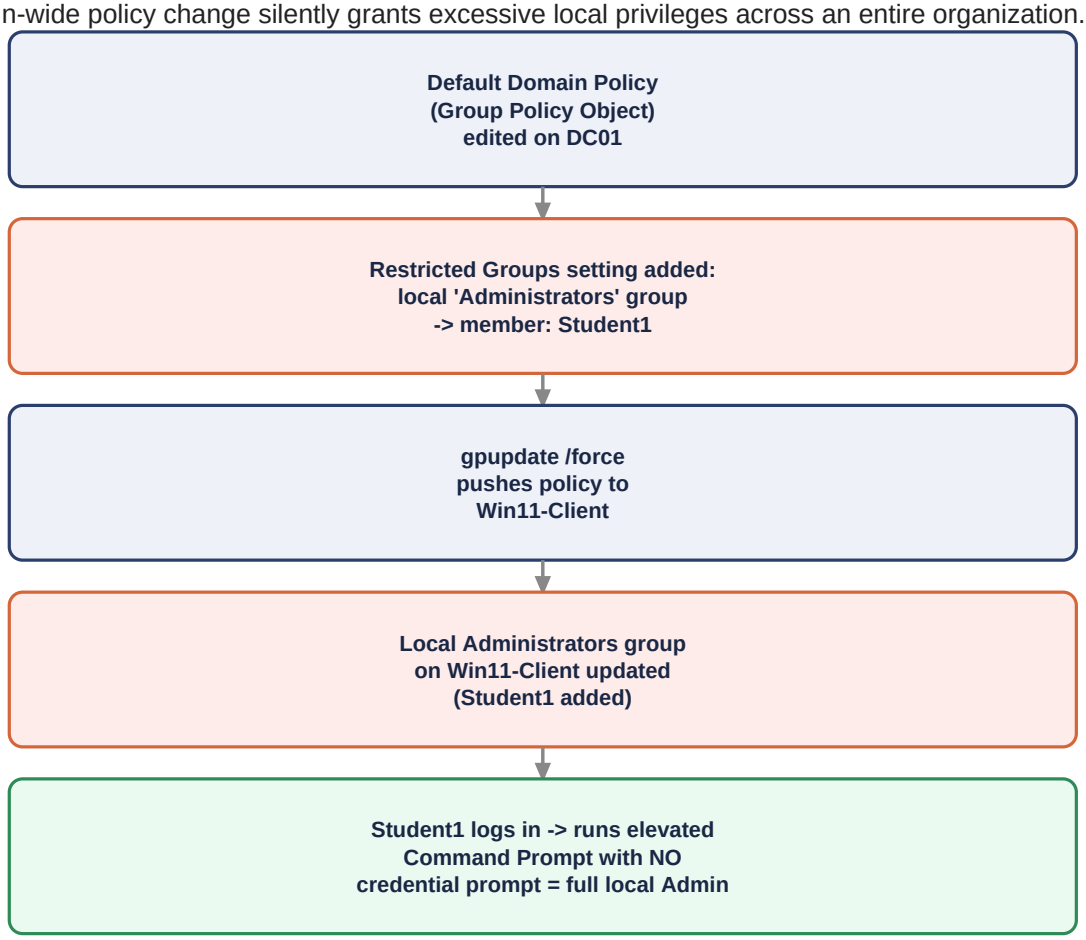


Figure 6 — How the misconfiguration propagates from the Domain Controller to full local Administrator access.

4.1 Misconfiguring the policy on the Domain Controller

Inside the Default Domain Policy, under `Computer Configuration` → `Security Settings` → `Restricted Groups`, the local 'Administrators' group was configured with Student1 as a member:

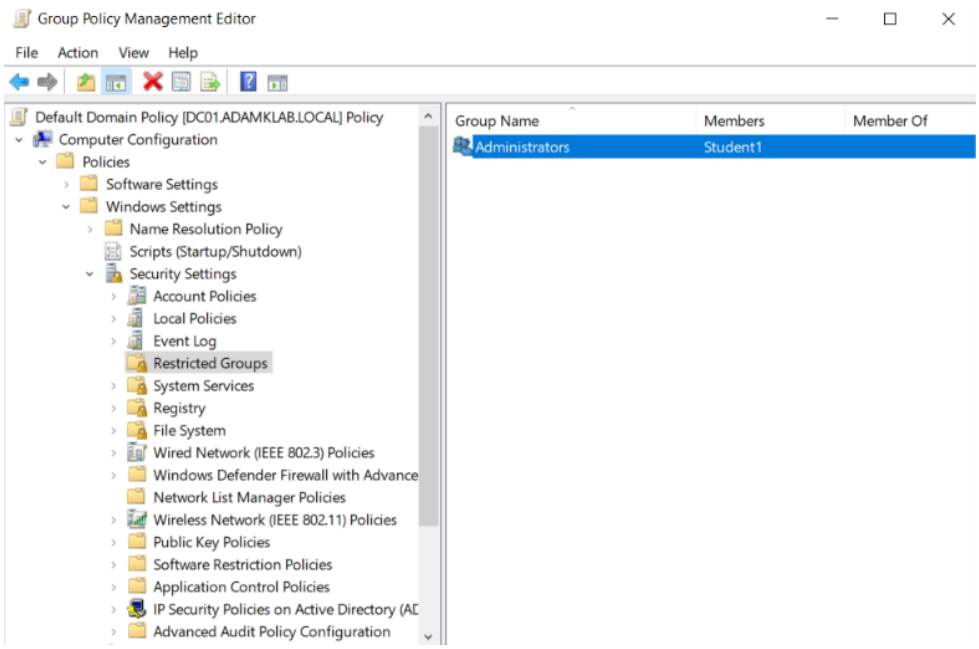


Figure 7 — Restricted Groups policy: Student1 configured as a member of the local Administrators group.

4.2 Applying the policy and confirming escalation on the workstation

The policy was pushed with `gpupdate /force`, and after a restart Student1's token reflected the new membership:

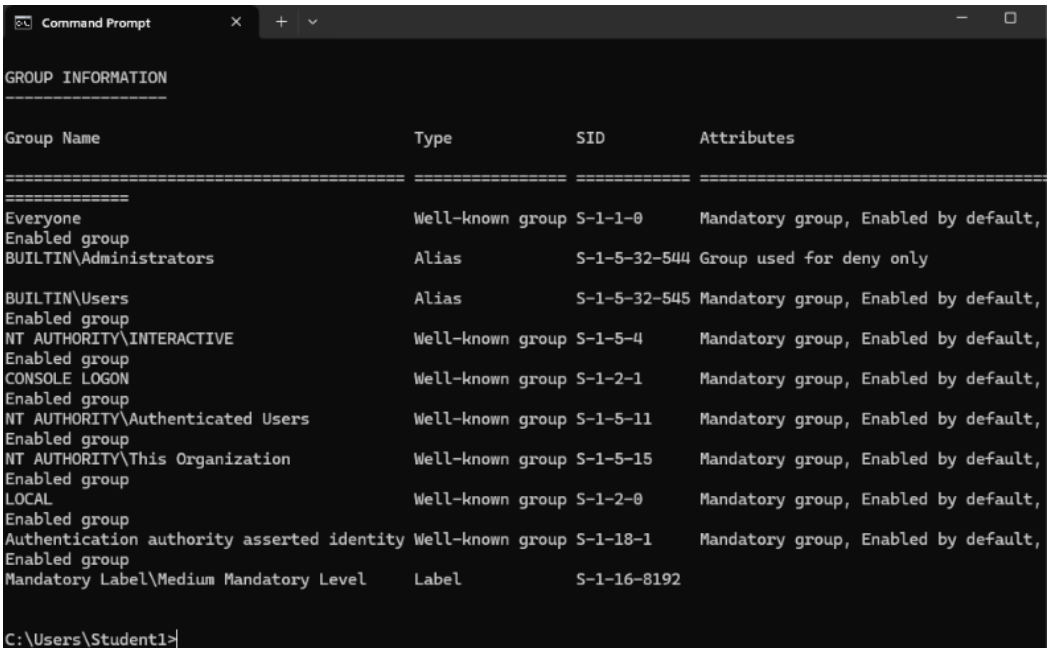


Figure 8 — whoami /groups as Student1 after the policy applied — now in BUILTIN\Administrators.

The decisive proof: launching Command Prompt as Administrator while logged in as Student1 succeeded with **no UAC credential prompt at all** — full local Administrator access was obtained purely through the policy misconfiguration, without using any privileged account's password.

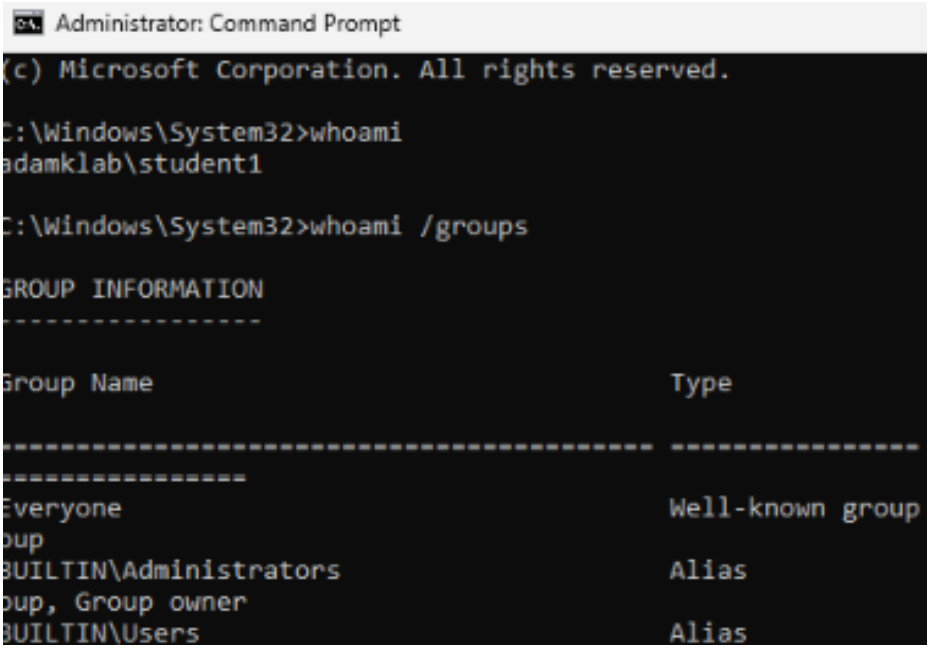


Figure 9 — Elevated 'Administrator: Command Prompt' opened directly as Student1 — privilege escalation confirmed.

Summary — Part A

- Active Directory domain 'adamklab.local' built and operational (DC01).
- Windows 11 workstation (Win11-Client) successfully joined to the domain.
- Standard user 'Student1' created with no administrative rights (baseline confirmed).
- A Group Policy 'Restricted Groups' misconfiguration was introduced and exploited, escalating Student1 to full local Administrator on the workstation — confirmed via whoami /groups and an elevated Command Prompt opened with zero credential prompts.